College of Business
香港城市大學
City University of Hong Kong

FinTech and AI in Finance

Class 8: DeFi, NFTs, and Real-World Assets

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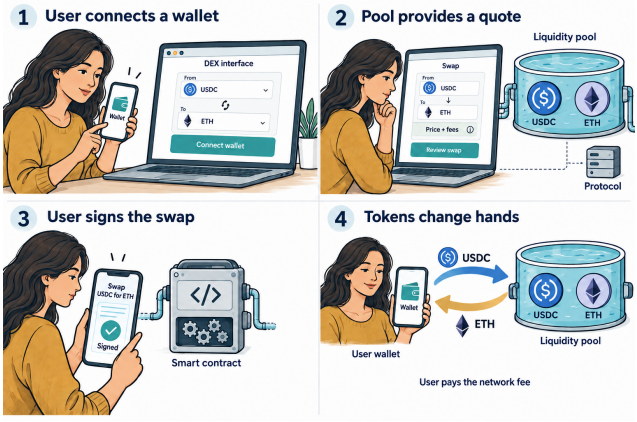
Semester A, 2026/27

City University of Hong Kong

Decentralized Finance

DeFi: A Token Swap

Decentralized finance (DeFi) runs financial services on public blockchains.



Source: Ethereum.org, DeFi. Simplified transaction.

Why Use DeFi When Banks and Exchanges Exist?

Consider someone who already holds assets in a blockchain wallet.

- **Trading.** Exchange one token for another directly from the wallet using a shared pool.
- **Borrowing.** Obtain spendable tokens against existing crypto holdings without first selling those holdings.
- **Access.** Compatible applications use the same assets and public transaction records.

Whether this is attractive depends on fees, available liquidity, and the risks the user takes.

Follow One Token Swap

A user holds USDC and wants ETH. The pool holds both tokens.

- **Check the quote.** The application shows the ETH expected for the USDC offered and the relevant fees.
- **Authorize the trade.** The user signs an instruction. The contract checks the spending permission and trade conditions.
- **Exchange the assets.** If the trade executes, USDC enters the pool and ETH goes to the user. The ledger records the new balances.

Pool reserves affect the execution price, as in the Class 7 example.

What TVL Measures

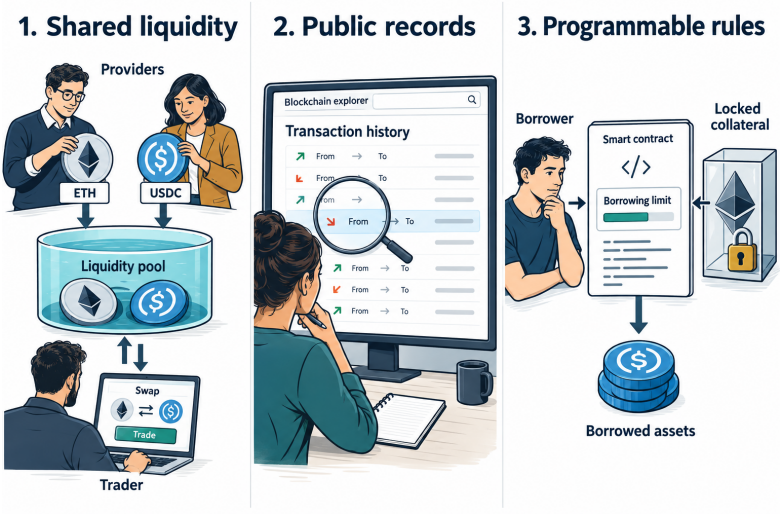


Source: DefiLlama historical TVL API, through 29 July 2026. Weekly endpoints shown. TVL reflects assets counted in protocol contracts and can move with token prices and coverage.

What the chart can and cannot show

TVL tracks assets counted in protocol contracts. It is useful for monitoring scale and drawdowns, but it is not protocol equity, revenue, user activity, or market liquidity.

Where DeFi Can Create Value

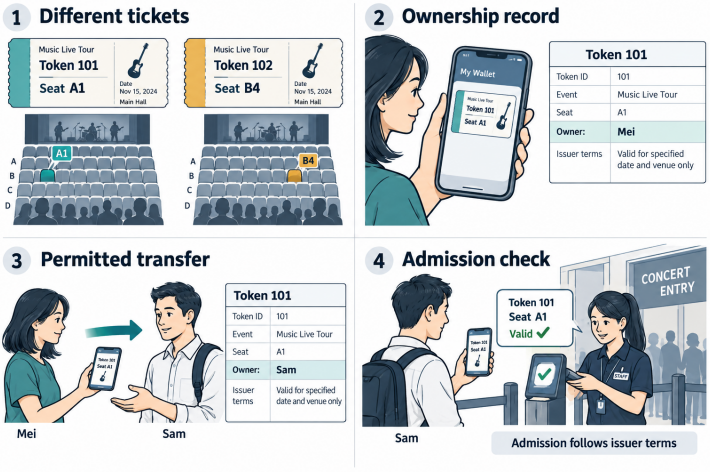


Disadvantages of DeFi

- **Can the transaction execute safely?** A contract exploit can move or lock assets. Congestion can delay action. Check audits, permissions, transaction costs, and recovery arrangements.
- **Can collateral be sold in time?** Price falls and leveraged positions can overwhelm liquidation. Check borrowing limits, price data, buyers, and whether yield depends on incentives.
- **Who can change the rules?** Voting power and emergency authority determine changes to parameters and treasury use. Check disclosure, delays, and accountability.

NFTs and Holder Rights

Non-Fungible Tokens (NFTs)



Illustrative ticket design. Admission and transfer follow the issuer's terms.

Applications of NFTs

Application	What the token identifies	What creates economic value
Digital art and collectibles	a specific edition or collection item	creator, scarcity, community, marketplace, and licence
Gaming	a character, item, land parcel, or access right	continuing game use and platform rules
Ticketing and membership	admission or a recurring benefit	issuer recognition, transfer policy, and service delivery
Identity or credentials	a claim about a person or qualification	trusted issuer, privacy, revocation, and acceptance
NFT finance	collateral offered against a loan	appraisal, liquidity, custody, and liquidation

The token identifies the claim; the issuer or ecosystem makes the claim useful.

An NFT Connects a Token, Metadata, and a Claim

Layer	What is recorded	What can fail
Token	owner address and transfer history	stolen key, bad contract, or blocked transfer
Metadata and media	a link to the image, video, ticket, or game item	broken storage link or changed metadata
Legal or platform rights	copyright licence, admission, benefits, or redemption terms	unclear contract or platform refusal

Owning the token does not automatically mean owning the copyright or the underlying asset.

Challenges of NFTs

- **Can you sell it?** A quoted NFT price needs a willing buyer. Check trading activity, ownership concentration, and executable bids.
- **Can you access and transfer it?** Phishing, stolen keys, or bad approvals can cause losses. A failed marketplace or broken media link can disrupt use. Check permissions, storage, and recovery options.
- **What must the issuer deliver?** Read the licence and platform terms. Identify who provides the promised service and how the holder can enforce that right.

NFT Uses and Holder Rights

- **Market cycle:** the 2021 collectibles boom was followed by a sharp correction in 2022–23.
- **2024–26:** experimentation continues in membership, ticketing, gaming, identity, and asset-linked claims.
- **Holder rights:** value depends on the benefit the token grants and whether its issuer can deliver it.

Tokenized Real-World Assets

Real-World Assets (RWAs): Tokenized Bonds

Tokens record claims on assets. Legal terms determine investors' rights.



Illustrative custody structure. Source: BIS Bulletin 72 (2023).

A Bond before and after Tokenization

Illustrative bond: an investor lends $1,000 for one year at a 5% coupon.

- **Cash flows.** The issuer promises $50 interest and repayment of $1,000 at maturity.
- **Records.** Tokenization changes how holdings and transfers are recorded and can connect them with payment instructions.
- **Investor rights.** The contract determines who owes the money and how the claim is enforced. The issuer can still default.

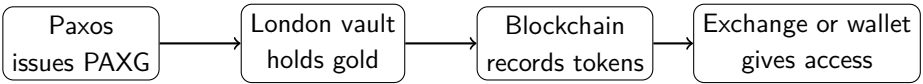
A more convenient ownership record does not change the promised cash flows in this example.

Paxos: Small Investments in Physical Gold

- Investors may want allocated gold without buying, storing, and transferring a large bar. Physical delivery is cumbersome for small purchases.
- Paxos issues PAX Gold (PAXG). Each token represents one fine troy ounce of allocated gold, with fractional holdings possible.
- The investor still needs a custodian and a redemption arrangement. The token changes transfer and access, while the gold stays in a vault.

Source: Paxos, PAX Gold product information. Checked 13 October 2026.

Paxos: Gold Custody and Token Transfer



- Paxos provides gold-allocation lookup for eligible on-chain wallets and publishes monthly attestation reports.
- Transferring PAXG changes the holder. Physical redemption also requires the issuer's eligibility, minimum-size, and delivery procedures.

Source: Paxos PAX Gold. Roles simplified.

Multiple-Choice Question: Can a Token Verify the Warehouse?

An immutable token represents warehouse inventory, but no independent audit prevents the same goods from being pledged twice. Which statement is best?

A) Tokenization removes custody risk.

B) The blockchain verifies physical inventory automatically.

C) Off-chain verification and legal-claim risk remain.

D) Token uniqueness guarantees ownership title.

Discuss
Separate the token's on-chain history from custody and enforceable rights over the physical asset.

Paxos: Fees and the Limits of Tokenized Gold

- Paxos currently charges creation and redemption fees under its published schedule, with no customer gold-storage fee.
- The token makes small holdings and transfers easier. Converting to physical gold remains a different service with separate conditions and costs.
- For a customer, the comparison is total cost and exit liquidity versus a gold ETF or bullion dealer. More token transfers do not prove better liquidity.

Sources: Paxos fee schedule, updated 19 August 2026; product information. Checked 13 October 2026.

Three Ways to Attach a Token to a Real Asset

- **Direct ownership.** The official register or recognised custodian records ownership of the asset. Token transfers must agree with that record. Custody and transfer restrictions still matter.
- **A company or trust owns the asset.** The investor holds a claim on that entity. Legal documents determine cash flows and asset protection. Governance, valuation, and servicing affect the claim.
- **A price-linked contract.** A counterparty promises a payoff linked to the asset's price. No ownership of the reference asset transfers. Payment depends on the contract, collateral, and counterparty.

For each structure, identify the investor's claim and who must honour it.

ChinaAMC: What Does a Tokenized Fund Hold?

ChinaAMC (Hong Kong): HKD and USD Digital Money Market Funds



- The token represents an interest in the fund. Its return comes from the portfolio; the share value can change.
- The manager earns management fees. Tokenization changes how shares are recorded and transferred, not how interest is earned.

Economic flows; return can accrue in NAV. Source: ChinaAMC fund information, accessed 28 October 2026.

HSBC Orion: The Issuer and Investor Need

- An issuer needs to distribute bonds and record ownership. Investors need reliable delivery, payment, and later interest payments.
- Conventional issuance connects several record keepers. Reconciling their records and coordinating settlement add work and delay.
- HSBC Orion offers a digital issuance and settlement platform. The government or company issuing the bond still owes the principal and interest.

Sources: HKSAR Government, November 2025; HSBC, February 2026.

ChinaAMC: Redeem the Fund or Sell the Token?

26 October 2026: 24/7 secondary trading began for the HKD and USD funds' Class F tokenized shares on PantherTrade.

Redeem in the primary market

Investor $\longrightarrow$ distributor $\longrightarrow$ fund

At the applicable net asset value (NAV), with dealing deadlines and a payment schedule.

Sell in the secondary market

Seller $\longleftrightarrow$ PantherTrade $\longleftrightarrow$ buyer

At the available market bid, which can differ from NAV even when fund dealing is closed.

Hypothetical weekend sale: reference NAV HK$100; bid HK$99.80.

1,000 shares $\times$ HK$99.80 $=$ HK$99,800: **HK$200 below NAV (20 bp)**, before fees.

24/7 trading is not 24/7 redemption at NAV.

Sources: ChinaAMC announcement, 23 October 2026; SFC, 26 October 2026.

Multiple-Choice Question: TVL and New Capital

A user locks $100 of collateral in Protocol A and borrows $60 of newly issued stablecoins against it. The user deposits those coins in Protocol B. Both the collateral in A and the deposit in B count as TVL.

What are the combined reported TVL and the user's original outside capital?

A) TVL of $160 and original outside capital of $100.

B) TVL of $160 and original outside capital of $160.

C) TVL of $100 and original outside capital of $160.

D) TVL of $60 and original outside capital of $100.

Multiple-Choice Question: Transferability and Market Liquidity

A platform allows transfers of tokenized private assets at any time, but investors cannot find buyers near the quoted price.

Which change most directly addresses the exit-liquidity problem?

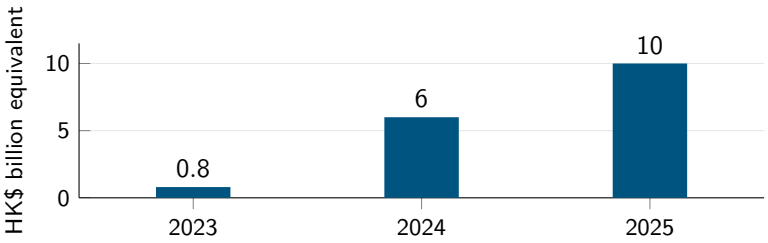
A) Divide each token into more units while leaving demand unchanged.

B) Record the same ownership information on an additional blockchain.

C) Attract funded buyers and market makers willing to quote executable prices.

D) Replace the trading interface without changing the available counterparties.

HKSAR Digital Green Bonds: Three Issuances



- **Government debt:** The HKSAR Government owes the interest and principal.
- **Green proceeds:** Funds finance or refinance eligible green projects.
- **Wider issuance:** The 2024 and 2025 offerings each covered HKD, RMB, USD, and EUR.

Sources: HKMA announcements for 2023, 2024, and 2025. HK$ equivalents rounded.

HSBC Orion: Bond Delivery against Payment

- For the HKSAR Government's 2025 digital green bonds, Orion provided the platform for issuance and settlement.
- Delivery versus payment links the bond transfer to its payment. The HKD and RMB tranches offered e-HKD and e-CNY settlement options alongside conventional arrangements.
- The earlier 2023 pilot used Goldman Sachs' GS DAP, with primary settlement on T+1. Different platforms can support the same financial obligation.

Sources: HKMA announcements, 16 February 2023 and 11 November 2025.

HKSAR Digital Green Bonds: Interest and Principal

Illustrative holding using the 2023 terms: HK$1 million face value, a one-year maturity, and a 4.05% coupon. Assume all payments are made in full.

$$\text{Interest} = 1{,}000{,}000 \times 4.05\% = \boxed{\text{HK\$40,500}}$$

$$\text{Principal plus interest} = 1{,}000{,}000 + 40{,}500 = \text{HK\$1,040,500}$$

- Tokenization changes how ownership and settlement are recorded. The government remains responsible for payment.
- "Green" describes the use of proceeds. Bondholders hold government debt, rather than ownership of individual green projects.

Source: HKMA, 16 February 2023. Holding size is illustrative.

Short-Answer Question: Evaluating a Tokenized Green Bond

Consider a tokenized HKSAR government green bond with the same coupon and maturity as a conventional bond. The platform links delivery to payment and shares transaction records.

1. Explain one potential saving in settlement or reconciliation. How could a bank measure the saving?

2. Who owes the interest and principal? Explain whether bondholders own the green projects financed by the bond.

HKSAR Is Moving from Pilots to Real-Value Transactions

- HKMA launched the Project Ensemble Sandbox in August 2024 to test tokenized deposits and digital assets.
- EnsembleTX began a controlled pilot in November 2025 for real-value transactions using tokenized HKD deposits and digital assets.
- The policy direction is toward interoperable settlement, not a promise that every token will be liquid or available to retail investors.
- Each transaction still requires a legal claim, permitted participants, money settlement, controls, and failure recovery.

Sources: HKMA, Project Ensemble and 2026 Fintech Promotion Blueprint.

China's Securities Regulator and Tokenized Assets

- The **China Securities Regulatory Commission (CSRC)** regulates securities markets. Its February 2026 guidelines cover offshore asset-backed security tokens supported by assets in China.
- **Illustrative proposal:** a firm pools rental receivables in China and offers tokens to overseas investors. Under the guidelines, the domestic entity controlling the assets must file with the CSRC.
- Investors need to know who owes them cash and what assets support the claim. The issuer must meet the applicable financing and authorization rules, even when a blockchain records ownership.

Sources: CSRC Announcement [2026] No. 1; CSRC policy explanation. Rental-receivables proposal is hypothetical.

HSBC Orion: Commercial Progress and Adoption

- HSBC won the UK government's DIGIT platform mandate in February 2026. A mandate for a pilot is different from completed issuance.
- The commercial opportunity is issuance and securities servicing for institutional clients. A large bond amount is the issuer's financing, not platform revenue.
- Adoption depends on investor access, compatible settlement arrangements, and lower operating costs. A new ledger alone cannot create secondary-market buyers.

Source: HSBC, DIGIT mandate, 12 February 2026. Business implications are classroom analysis.

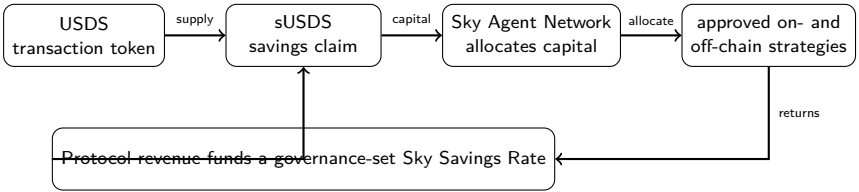
DeFi Lending: Sky and Aave

Case 1: Sky — the System Formerly Called MakerDAO

- **The user.** USDS serves transfers, while sUSDS provides a variable savings return. Compare the claim and redemption route before choosing between them.
- **The source of income.** Protocol capital goes into loans and approved strategies. Wallets, exchanges, and other services connect users to these arrangements.
- **The operating decisions.** SKY governance sets rates, agents, limits, and upgrades. Liquidity and operating history matter when users withdraw or a counterparty fails.

MakerDAO rebranded its ecosystem as Sky in 2024. Source: Sky ecosystem documentation, accessed July 2026.

Sky's 2026 Product Stack



USDS is the transactional entry point; sUSDS passes a governance-set share of protocol revenue to savers. The interface, protocol, agents, and off-chain counterparties are distinct roles.

Source: Sky.money USDS and sUSDS documentation, June 2026; company material is not a guarantee of yield or safety.

Where Sky's Yield Comes From, and Who Bears the Risk

- **Trace the income.** Loans and approved on- or off-chain strategies generate returns. Separate earned revenue from incentives or subsidies used to attract users.
- **Trace a loss.** Collateral, excess asset value, liquidation, and governance actions affect who bears it. Check how losses could reach USDS or sUSDS holders and their redemption rights.
- **Identify the decision makers.** Who selects agents and counterparties, sets limits, and pays distribution incentives? These choices connect asset risk with the return offered to savers.

The savings rate allocates risky protocol income. It is not a bank-guaranteed deposit rate.

Worked Example: Funding a Savings Yield

Suppose a protocol earns \$6m from reserve assets and \$4m from borrower fees in one year. Expenses are \$2m. Savings balances remain \$200m throughout the year.

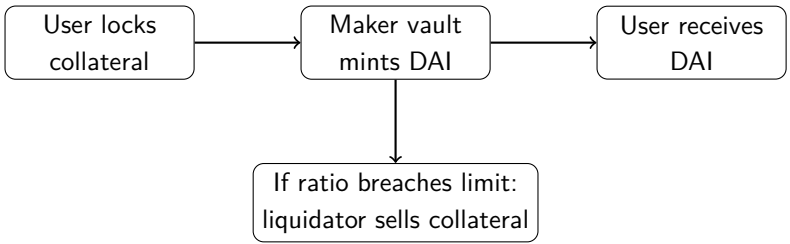
If all net income goes to savers, with no subsidies:

$$\text{Distributable income} = 6 + 4 - 2 = \$8\text{m}$$

$$\text{Savings yield} = \frac{8}{200} = 4\%$$

A 5% payout would require \$10m. The extra \$2m must come from additional earnings, reserves, or a subsidy.

One Legacy Mechanism Still Matters: Vault Liquidation



- Over-collateralization protects DAI only if prices, oracles, and liquidation markets keep working.
- RWA collateral adds custodians and enforceable off-chain claims to this flow.

Worked Example: A Liquidation Threshold

A vault holds \$150 of collateral against \$100 of debt. Liquidation becomes possible below a collateral-to-debt ratio of 125%.

The collateral price falls 20%, while debt stays fixed.

$$\text{New collateral value} = \$150 \times (1 - 0.20) = \$120$$

$$\text{New collateral-to-debt ratio} = \frac{120}{100} = 120\%$$

- The ratio falls below 125%, so the vault becomes eligible for liquidation.
- Liquidation can begin while collateral still exceeds the debt.

DAO Voting: Who Can Change the Rules?

A **DAO** coordinates decisions through governance rules, often using token-weighted votes. Proposals may change collateral limits or allocate treasury funds.

40 holders oppose

1 voting unit each: **40 votes**

1 holder supports

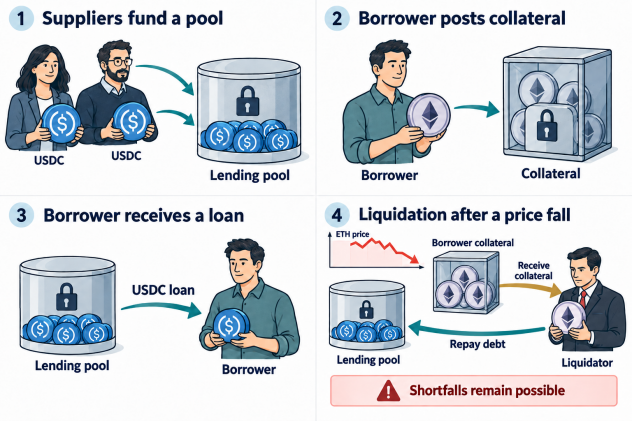
60 voting units: **60 votes**

- Under a simple majority rule, voting power can outweigh the number of people.
- A lending claim or savings balance does not by itself confer governance votes or legal equity.

Hypothetical vote, assuming all quorum and proposal requirements are met. Actual token rights vary. Source: Aave governance documentation.

Aave: Collateralized Borrowing

Suppliers earn variable interest. Borrowers pay interest on their loans.



Sources: Aave, Health Factor & Liquidations; Aave App Disclosures.

Short-Answer Question: Collateral Value and Loan Losses

A lending platform makes a \$100 loan against collateral worth \$150. Before the collateral can be sold, its price falls by 40%. Selling costs are \$5. Ignore interest.

1. Calculate the net cash recovered by selling the collateral.
2. Calculate the loan shortfall if there is no further recovery.
3. Explain why recording the collateral on a blockchain does not prevent this loss.

Aave Example: Selling ETH or Borrowing USDC

Alice owns 10 ETH at $3,000 each and wants 15,000 USDC.

Immediately after the transaction	**Sell 5 ETH**	**Borrow**
ETH retained	5	10 (collateral)
USDC held	15,000	15,000
USDC debt	0	15,000
Net wealth at initial prices	$30,000	$30,000

Borrowing retains exposure to all 10 ETH, but adds interest and liquidation risk. The pool transfers existing USDC supplied by lenders.

Hypothetical Aave-style loan. Assume 1 USDC = $1. Cash remains unspent in both columns. Ignore interest, gas, and other positions in the balance-sheet comparison.

Borrowing Limit and Liquidation Threshold

Assume maximum loan-to-value (LTV) is **60%**, but the liquidation threshold is **80%**. Alice initially borrows 15,000 of a possible 18,000 USDC.

$$\text{Health factor (HF)} = \frac{\text{Collateral value} \times 80\%}{\text{Debt value}}$$

ETH price ($)	Collateral ($)	Debt (USDC)	HF
3,000	30,000	15,000	1.60
2,000	20,000	15,000	1.067
1,800	18,000	15,000	0.96

At $2,000, LTV is 75%: no additional borrowing, but HF remains above 1. Liquidation becomes possible when HF falls below 1.

Teaching parameters. Same loan and assumptions as the previous page. Aave, Health Factor & Liquidations.

Liquidation: Who Pays and Who Receives ETH?

ETH falls to \$1,800. Suppose the rules allow a liquidator to repay 6,000 USDC of Alice's debt with a **5% bonus**.

$$\text{ETH received} = \frac{6,000 \times 1.05}{1,800} = \boxed{3.5 \text{ ETH}}$$

- Alice retains 6.5 ETH as collateral and owes 9,000 USDC.
- Collateral equity falls from \$3,000 to \$2,700. The \$300 difference pays the liquidation bonus.
- The new HF is $6.5 \times 1,800 \times 80\%/9,000 = 1.04$.

Hypothetical allowed repayment and bonus. Price stays at \$1,800 during execution. Ignore accrued interest, gas, price impact, and any protocol share of the bonus. Previously borrowed cash is unchanged.

Why a 5% Borrowing Rate Can Pay Suppliers 2.7%

Suppose lenders supply 1m USDC. Borrowers, including Alice, draw 600,000. The **utilization rate** is 60%.

At a 5% annual borrowing rate and a 10% reserve factor:

$$\begin{aligned}
\text{Borrower interest} &= 600,000 \times 5\% = 30,000,\\
\text{Treasury share} &= 30,000 \times 10\% = 3,000,\\
\text{Supplier rate} &= \frac{30,000 - 3,000}{1,000,000} = \boxed{2.7\%}.
\end{aligned}$$

400,000 USDC remains available for withdrawals. Treasury income is before incentives and operating costs.

Hypothetical one-year simple-interest example with constant balances and rates. No losses or other income. Aave, supplying; reserve factor.

Multiple-Choice Question: Withdrawals from a Lending Pool

A pool has $100m of deposits, of which $90m is lent out and $10m remains available. Suppliers request $30m immediately. There are no new deposits or loan repayments.

Which conclusion follows?

A) Adequate borrower collateral makes all $30m immediately available.

B) Only $10m is available, even if the loans remain fully recoverable.

C) The withdrawal request proves the pool has a $20m credit loss.

D) Raising borrowing rates instantly supplies the missing $20m.

Case 2: Aave Ecosystem and Governance

- **Access and products.** Wallets and services that compare protocols connect users to Aave. GHO, an Aave-governed stablecoin, extends its product range.
- **Loan rules.** Token governance sets eligible assets, collateral parameters, risk limits, incentives, and upgrades. These decisions affect how much a customer can borrow.
- **Operating across networks.** More networks expand access but add systems to monitor, including price feeds, bridges, and liquidation arrangements. A failure in these services can affect repayment and withdrawals.

Two DeFi Founders Who Changed Their Original Designs

Person	First bet	What changed	Lesson
 **Rune Christensen**	MakerDAO used over-collateralized vaults to create DAI	the system evolved into Sky, with USDS, savings, agents, and more off-chain assets	governance must adapt the balance sheet
 **Stani Kulechov**	ETHLend matched borrowers and lenders peer to peer	the team rebuilt it as Aave around pooled liquidity when bilateral matching did not scale	a new market needs a scalable matching design

The interesting story is the redesign

Both founders kept the financial job but changed the mechanism: liquidity design, risk limits, and governance mattered as much as code.

Sources: MakerDAO/Sky ecosystem history; Aave Labs history and 2026 contributions report. Portraits are AI-generated editorial illustrations, not photographs.

Yield and Risk: Aave and Sky

- **Borrower interest: Aave.** USDC suppliers earn interest paid by USDC borrowers. Lower borrowing demand can reduce supplier returns.
- **Protocol income: Sky.** Agents pay for capital they deploy in lending and other strategies. Protocol revenue funds the governance-set sUSDS savings rate.
- **Collateral losses: Aave.** ETH backing a USDC loan can fall sharply. Liquidation limits losses, but the proceeds may not cover the full debt.

Sources: Aave, supplying; Aave, liquidations; Sky, sources of yield.

A Cross-Chain Claim on 5 ETH

Suppose a bridge uses a **lock-and-mint** design to move ETH exposure to another blockchain.

1. Lock **5 ETH** in a contract on Ethereum.
2. After verification, mint **5 claim tokens** on the destination chain. The original ETH remains locked.
3. To return, burn those claim tokens. After verification, release the original **5 ETH**.

Who verifies each step? If the locked ETH is stolen, the claim token may lose its backing even if the wallet still shows a balance.

One illustrative bridge design, ignoring fees. A bridged claim depends on the bridge's backing and verification. Source: Ethereum.org, Bridges.

What Do These Tokens Give Their Holders?

- **Aave supply position:** supplied tokens plus accrued interest. Withdrawals depend on available pool liquidity.
- **The Sandbox LAND NFT:** a virtual plot for publishing or renting space for games. A tenant's game copyright remains with its creator.
- **HKSAR digital green bond on HSBC Orion:** the government owes coupon and principal. HSBC provides the digital assets platform.

Sources: Aave, withdrawals; The Sandbox, game rights; HKSAR Government, 11 November 2025.